ATLAS X FALCON-M:

- PROJECT ROADMAP
- PROJECT SPECIFICATIONS DOCUMENT
- ENGINEERING/DESIGNING LOG

What is Atlas?

Atlas will be a fully scratch-built radio controlled cargo aircraft, designed with a payload system to deploy midair

What is Falcon-M?

Falcon-M will be the payload: somewhat of a drone/lander hybrid: It will be ejected from Atlas midair and then will autonomously re-orient itself and land, much like a SpaceX reusable rocket booster.

In this document..

This document will include the project roadmap, Specifications, how the projects are unified, and will be used to document my progress in the form of logs.

Project Roadmap:

PHASE 00-Foundation

- **Both Atlas and Falcon-M**
  - Create a combined project roadmap.
  - Establish separate Atlas and Falcon-M repositories
  - Create a joint engineering specifications document.(This doc)
  - Establish a versioned CAD and simulation file structure.
  - Record existing Atlas progress and outstanding problems.
  - Create an engineering decision log.
- **Deliverable: A professional project structure with traceable progress.**

PHASE 01-ATLAS ENGINEERING

- **1A. Airframe**
  - Review existing CAD and 3D blockout.
  - Establish wingspan, length and wing geometry.
  - Calculate estimated empty mass.
  - Estimate centre of gravity and acceptable CG range.
  - Review structural loads and foamboard construction.
  - Review motor and propeller configuration.
- **1B. Flight systems**
  - Document radio receiver and transmitter configuration.
  - Establish servo requirements.
  - Create electrical architecture.
  - Establish battery and power requirements.
  - Plan failsafe behaviour.
- **1C. Payload interface**
  - Reserve a payload bay.
  - Investigate a mechanical retention mechanism.
  - Document payload mass and centre-of-gravity effects.
  - Establish a safe, ground-testable release interface.
- **Deliverable: Atlas Engineering Baseline v1.0, with drawings, mass estimates and electrical schematics.**

PHASE 02-Falcon-M conceptual development

- **2A. Vehicle geometry**

    - Investigate the rounded cylindrical/blended body shape.
    - Compare three-point and four-point landing arrangements.
    - Establish a preliminary internal component layout.
    - Determine how Falcon-M attaches to Atlas.
- **2B. Flight dynamics research**
    - Understand attitude, angular velocity and stability.
    - Investigate controlled orientation changes.
    - Compare electric VTOL propulsion architectures.
    - Investigate descent and landing control.
    - Investigate landing gear and suspension.
- **2C. Requirements**
    - Establish provisional mass and dimensional limits.
    - Define the landing-zone concept.
    - Define what happens if a landing attempt fails.
    - Identify the sensors and computing requirements.
- **Deliverable: Falcon-M Preliminary Design Review (PDR) document**.

PHASE 03-CAD and system integration

- **Both ATLAS and FALCON-M**
    - Complete Atlas's payload-bay geometry.
    - Create Falcon-M's initial 3D body.
    - Create a shared dimensional interface.
    - Check physical clearance and retention.
    - Evaluate the combined centre of gravity.
    - Produce assembly and exploded-view drawings.
    - Document the separation interface.

Deliverable: A complete digital assembly showing Atlas and Falcon-M together.

PHASE 04-Software and simulation

- **Both ALTAS and FALCON-M**
- **Atlas simulation**
- **Aerodynamic estimates, mass properties and performance analysis.**
- **Falcon-M simulation**
- **Attitude dynamics, controlled descent and virtual landing tests.**
- **Atlas software**
    - Aircraft mass-property calculations.
    - Aerodynamic and performance analysis.
    - Payload integration calculations.
    - Optional flight-data logging.
- **Falcon-M software**

    - 6-DOF rigid-body simulation.
    - IMU sensor model.
    - Virtual attitude-control system.
    - Propulsion and descent model.
    - Virtual landing-zone environment.
    - Simulation logs and performance graphs.
- **Joint software**
    - Simulate the carrier and payload as a combined system.
    - Model separation as a distinct event.
    - Record and visualise system performance.
- **Deliverable: Two documented simulation environments and a joint system simulation.**

PHASE 05-Physical prototypes

- **Atlas First**
    - Construct Atlas airframe.
    - Verify structure and electrical systems on the ground.
    - Conduct safe, supervised Atlas flight testing.
    - Build Falcon-M's non-flying structural prototype.
    - Test landing gear and suspension independently.
    - Test sensor and controller behaviour on a stationary rig.
    - Compare actual measurements against simulations.

Deliverable: Physical evidence of construction and testing, with documented improvements.

PHASE 06-Final engineering review

- **Submission**
    - Atlas final engineering report.
    - Falcon-M final engineering report.
    - Joint system architecture.
    - CAD and simulation demonstrations.
    - Bill of materials and expenditure records.
    - Test results, failures and design revisions.
    - Final demonstration video.

Deliverable: Two independently understandable Hack Club Fallen submissions that also tell one larger engineering story.